Form B

Core Focus on Proportions & Probability

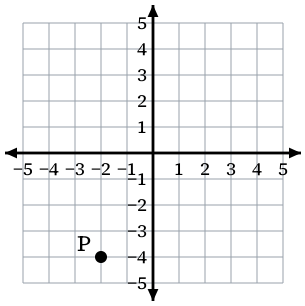
Block 5 Review ~ Direct Variation

ANSWER KEY

Name _____ Period _____ Date _____

Part I – Selected Response

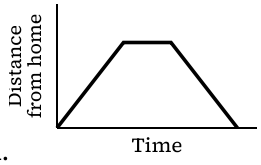
1. Which ordered pair represents the point graphed?

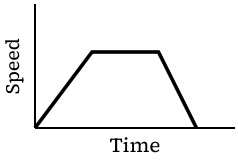


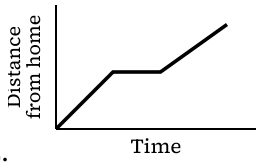
A. (−4, −2)
B. (−2, −4)
C. (2, 4)
D. (−2, 4)

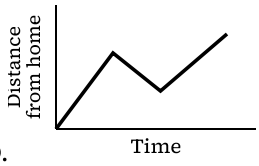
B. (−2, −4): start at the origin, move 2 left, then 4 down.

2. Which graph shows a person riding a bike away from home, stopping to rest, and then riding even farther away from home?

A. 

B. 

C. 

D. 

C. Distance from home goes up (riding away), stays flat (resting, distance does not change), then goes up again (riding farther away).

3. If the values in the table below model direct variation, what is the rate?

x	0	1	2	3	4
y	0	5	10	15	20

A. 5
B. $\frac{1}{5}$
C. 10
D. It does not model direct variation.

A. 5 (y ÷ x = 5 ÷ 1 = 10 ÷ 2 = 15 ÷ 3 = 20 ÷ 4 = 5)

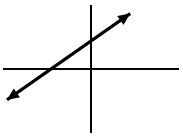
4. Which equation is the direct variation equation for the values in the table below?

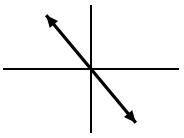
x	0	3	6	9	12
y	0	12	24	36	48

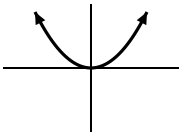
A. y = 12x
B. y = 3x
C. y = $\frac{1}{4}$x
D. y = 4x

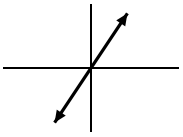
D. y = 4x (m = y ÷ x = 12 ÷ 3 = 4; check 48 ÷ 12 = 4)

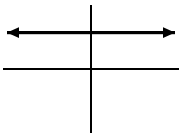
5. Which graphs show direct variation? Circle all that apply.

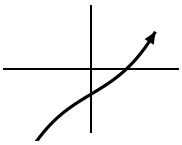
A. 

B. 

C. 

D. 

E. 

F. 

B, D. Each is a straight line through the origin. A, E and F miss the origin; C passes through the origin but is a curve.

6. Which equations are direct variation equations? Circle all that apply.

A. y = $\frac{x}{4}$
B. y = $\frac{4}{x}$
C. y = −6x
D. y = 2x + 1
E. y = 1.5x
F. y = $3x^2$

A, C, E. Each has the form y = mx (m = ¼, −6, 1.5). B has x in the denominator, D adds 1, F squares x.

7. What is the slope of the direct variation equation y = 0.25x?

A. $\frac{1}{4}$

B. 4

C. 25

D. $\frac{1}{25}$

A. ¼ (the slope is the number multiplying x: 0.25 = ¼)

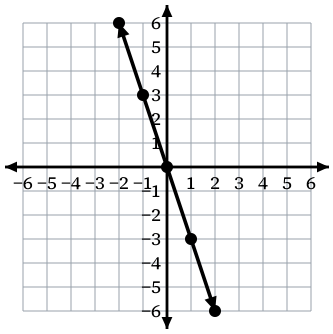
8. What is the direct variation equation for the line graphed below?

A. y = 3x

B. $y = -\frac{1}{3}x$

C. y = −3x

D. $y = \frac{1}{3}x$



C. y = −3x (the line passes through (0, 0) and (1, −3): m = −3 ÷ 1 = −3)

Form B

Core Focus on Proportions & Probability

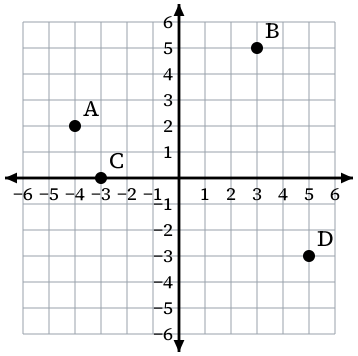
Block 5 Review ~ Direct Variation

ANSWER KEY

Name _____ Period _____ Date _____

Part II – Constructed Response

Write the ordered pair for each point on the coordinate plane to the right.



1. A

(−4, 2)

2. B

(3, 5)

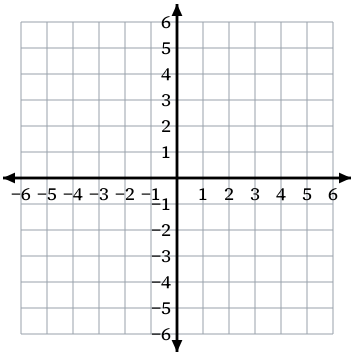
3. C

(−3, 0) — on the x-axis

4. D

(5, −3)

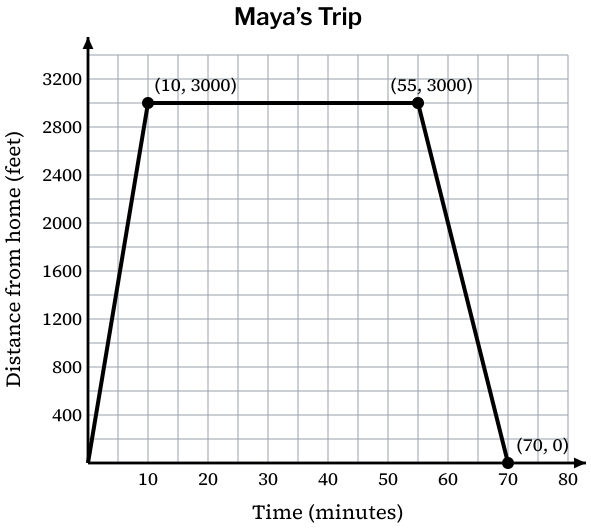
5. Plot the points (−3, 2), (3, 2), (3, −2) and (−3, −2) and connect them in order. Connect the last point listed to the first point. What shape do the points create?



Shape: _____

Rectangle. Points (−3, 2), (3, 2), (3, −2), (−3, −2): the top and bottom sides are 6 units long, the left and right sides are 4 units long, all corners are right angles.

6. Maya walked to the library, studied there, and then walked back home. The graph below shows her distance from home (feet) over time (minutes). For each part, explain how you know your answer is correct.



a. How far away is the library from Maya's house?

3,000 feet. The flat part of the graph is at 3,000 feet.

b. How long did Maya study at the library?

45 minutes. The graph is flat from 10 to 55 minutes: 55 − 10 = 45.

c. Find Maya's rate of speed in feet per minute for the walk to the library.

300 feet per minute. 3,000 ft ÷ 10 min = 300.

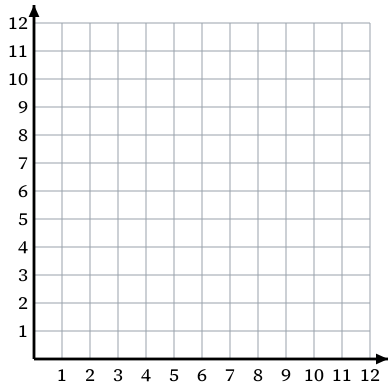
d. Find Maya's rate of speed in feet per minute for the walk home.

200 feet per minute. She walks 3,000 ft from minute 55 to minute 70: 3,000 ÷ 15 = 200.

Complete each input-output table and graph each function in Quadrant I. What is the slope of each direct variation equation?

7.

Input x	Function Rule y = 2x	Output y
0		
1		
2		
3		
4		

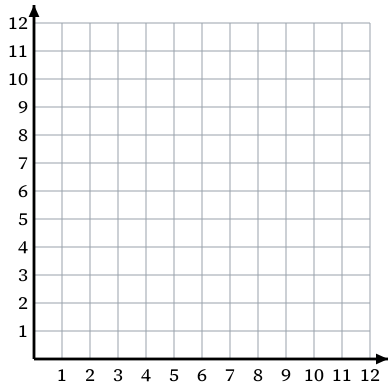


slope: _____

Outputs: 0, 2, 4, 6, 8 (2(0) = 0; 2(1) = 2; 2(2) = 4; 2(3) = 6; 2(4) = 8). Graph: points (0, 0), (1, 2), (2, 4), (3, 6), (4, 8) on a straight line through the origin. Slope = 2.

8.

Input x	Function Rule $y = \frac{1}{2}x$	Output y
0		
2		
4		
6		
8		



slope: _____

Outputs: 0, 1, 2, 3, 4 (½(0) = 0; ½(2) = 1; ½(4) = 2; ½(6) = 3; ½(8) = 4). Graph: points (0, 0), (2, 1), (4, 2), (6, 3), (8, 4) on a straight line through the origin. Slope = ½.

Determine whether each table models direct variation. Circle YES or NO. If it does, give the rate. If not, explain why not.

9.

x	y
−2	6
−1	3
0	0
1	−3
2	−6

direct variation? **YES** **NO**

rate/explanation:

YES; rate −3. y ÷ x = −3 for every pair except (0, 0): 6 ÷ (−2) = −3, 3 ÷ (−1) = −3, −3 ÷ 1 = −3, −6 ÷ 2 = −3.

10.

x	y
−2	−3
−1	−1
0	1
1	3
2	5

direct variation? **YES** **NO**

rate/explanation:

NO. When x = 0, y = 1, so the graph does not pass through the origin. The ratios y ÷ x are not equal (3 ÷ 1 = 3 but 5 ÷ 2 = 2.5).

Determine whether each equation models direct variation. Circle YES or NO. If it does, give the rate. If not, explain why not.

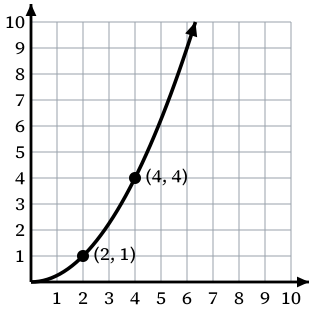
11. y = 0.4x direct variation? **YES** **NO** rate/explanation:

YES; rate 0.4. It has the form y = mx with m = 0.4.

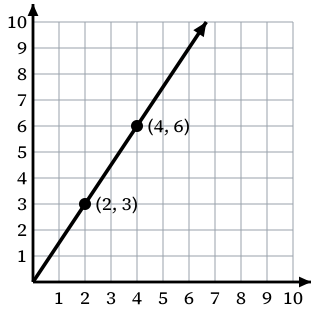
12. $y = 5x - 2$ direct variation? **YES** **NO** rate/explanation:

NO. The "− 2" means the line crosses the y-axis at (0, −2), not at the origin.

13. Ana says Graph B models direct variation and has a rate of $\frac{3}{2}$. Liam says Graph A models direct variation because it passes through the origin. Which person is correct? Explain your reasoning.



Graph A



Graph B

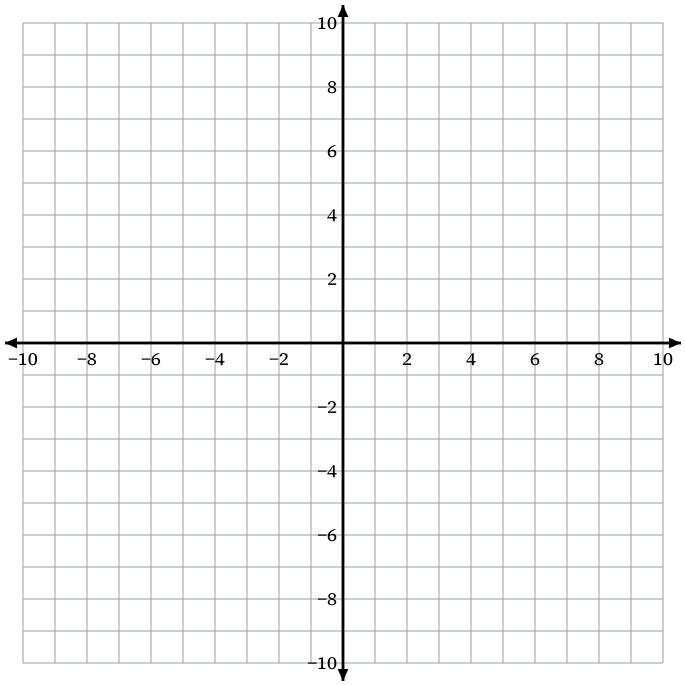
Ana is correct. Graph B is a straight line through (0, 0), and y ÷ x = 3 ÷ 2 = 6 ÷ 4 = 3/2. Graph A passes through the origin but is a curve, not a straight line (1 ÷ 2 = ½ but 4 ÷ 4 = 1), so it is not direct variation.

14. A printer prints 12 pages every minute. The table below shows the number of pages printed for the number of minutes the printer runs. Write an equation that shows the number of pages, y, printed in x minutes.

x	y
0	0
1	12
2	24
3	36
4	48

y = 12x (m = y ÷ x = 12 ÷ 1 = 24 ÷ 2 = 12)

15. Graph the direct variation equation $y = -2x$.



A straight line through the origin that falls from left to right with slope −2: it passes through (−5, 10), (−2, 4), (0, 0), (2, −4) and (5, −10), in Quadrants II and IV.

Core Focus on Proportions & Probability

Block 5 Review ~ Direct Variation

ANSWER KEY

Name _____ Period _____ Date _____

Part III – Problem Solving

The table below shows the amount of money Jada earned selling bracelets she made.

Bracelets Sold	Money Earned ($)
0	0.00
3	13.50
6	27.00
9	40.50
12	54.00

How many bracelets will Jada need to sell to earn $81.00?

18 bracelets. Rate = 13.50 ÷ 3 = $4.50 per bracelet (27 ÷ 6 = 4.50 too), so y = 4.5x. 81 = 4.5x → x = 81 ÷ 4.5 = 18. Check: 18 × 4.50 = 81.00.